

Active Beauty
Endothelyol™
Control of redness and rosacea

Crafted by white technology

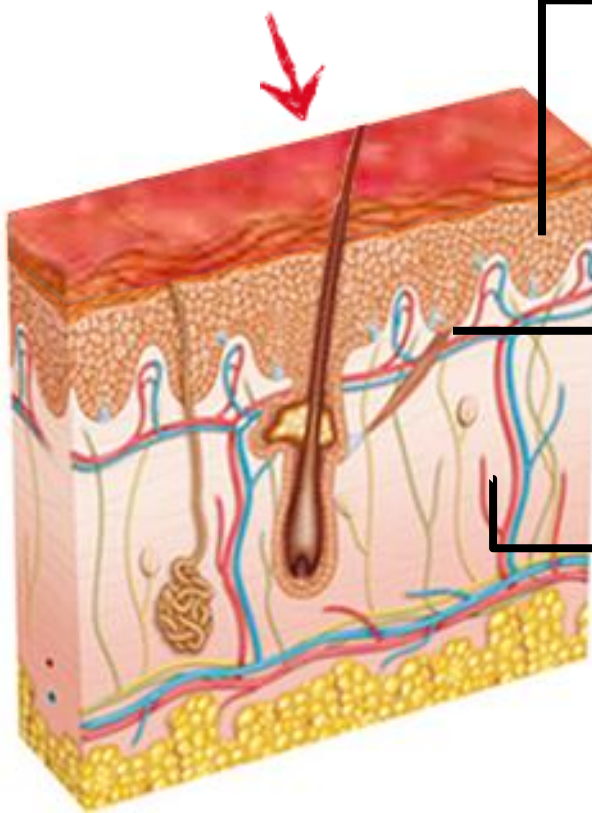


Givaudan

engage your senses

Mechanism of skin redness and rosacea

Mechanical or infectious stress
Internal deregulation



Keratinocytes

- IL8 and TNF- α secretion
 - ✓ (pain and inflammation)
- PGE2 release
 - ✓ Vasodilation (skin hyperreactivity)

Mastocytes

- Histamine release
 - ✓ Itching

Fibroblasts

- VEGF production
 - ✓ Neovascularisation

Redness and rosacea

A 5 in 1 global control of inflammation to reduce skin redness and rosacea

Endothelyol® is inspired by pharma to regulate all the key factors of redness and rosacea:

- ✓ downregulation of inflammation factors
- ✓ control of neovascularisation
- ✓ uniformization of skin tone
- ✓ improvement of rosacea in 28 days

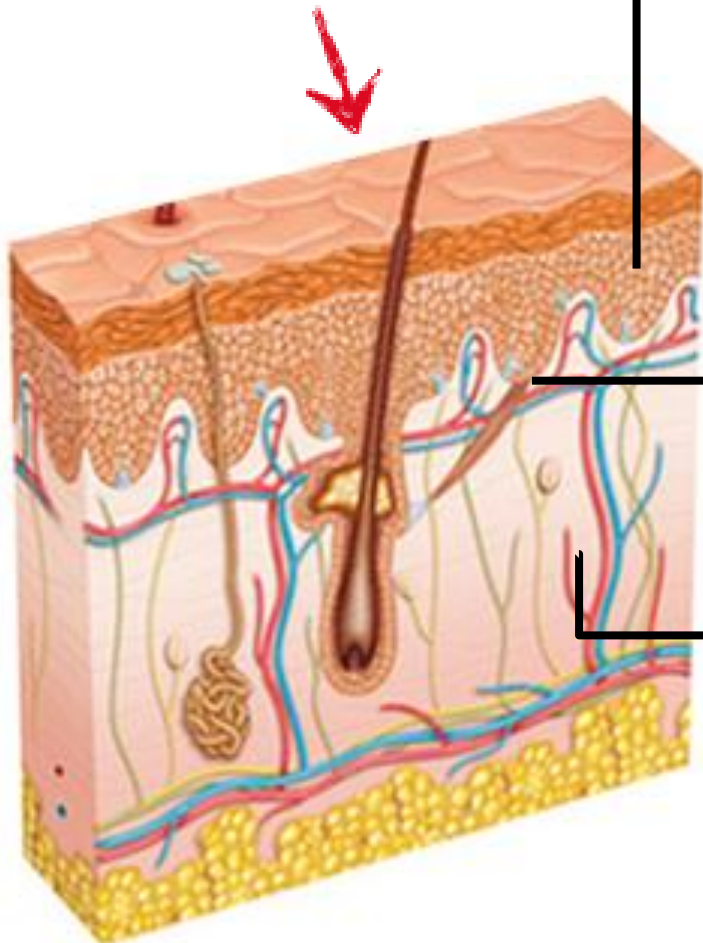


Endothelyol® is a pure active ingredient obtained by enzymatic modification of natural plants hydroxybenzoic acid and hydroxycinnamic acids, by means of a proprietary biotechnology process.

A 5 in 1 global control of inflammation to reduce skin redness and rosacea

Endothelyol™

Mechanical or infectious stress
Internal deregulation



Keratinocytes

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in vitro evaluations



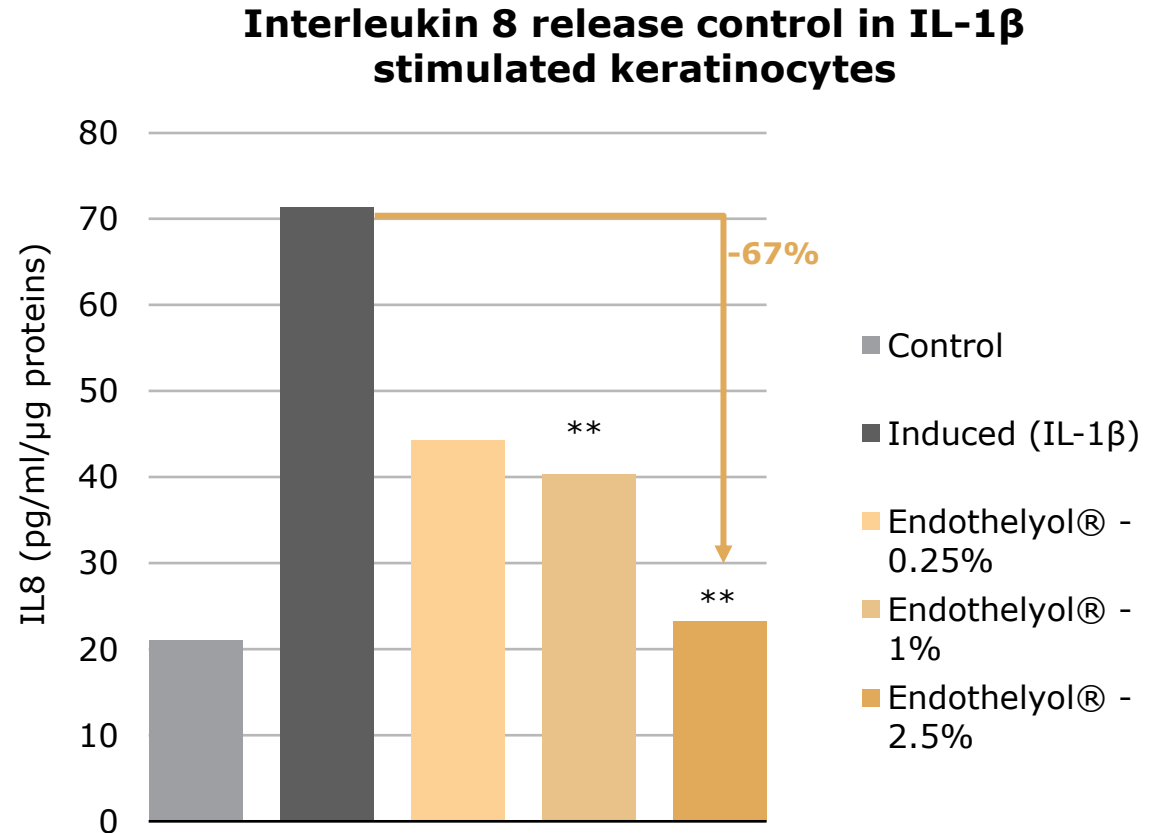
Tight control of redness factors

Interleukin 8 release control in IL-1 β stimulated keratinocytes

PROTOCOL

Endothelyol® was tested for its ability to control redness factors. ELISA quantification of IL8 released was done on keratinocytes extracted from clinical human skin sample incubated with the product for 48h.

→ Endothelyol®
downregulates IL-8
release by -67%.



**p<0.01 compared to stimulated control, Student's t Test

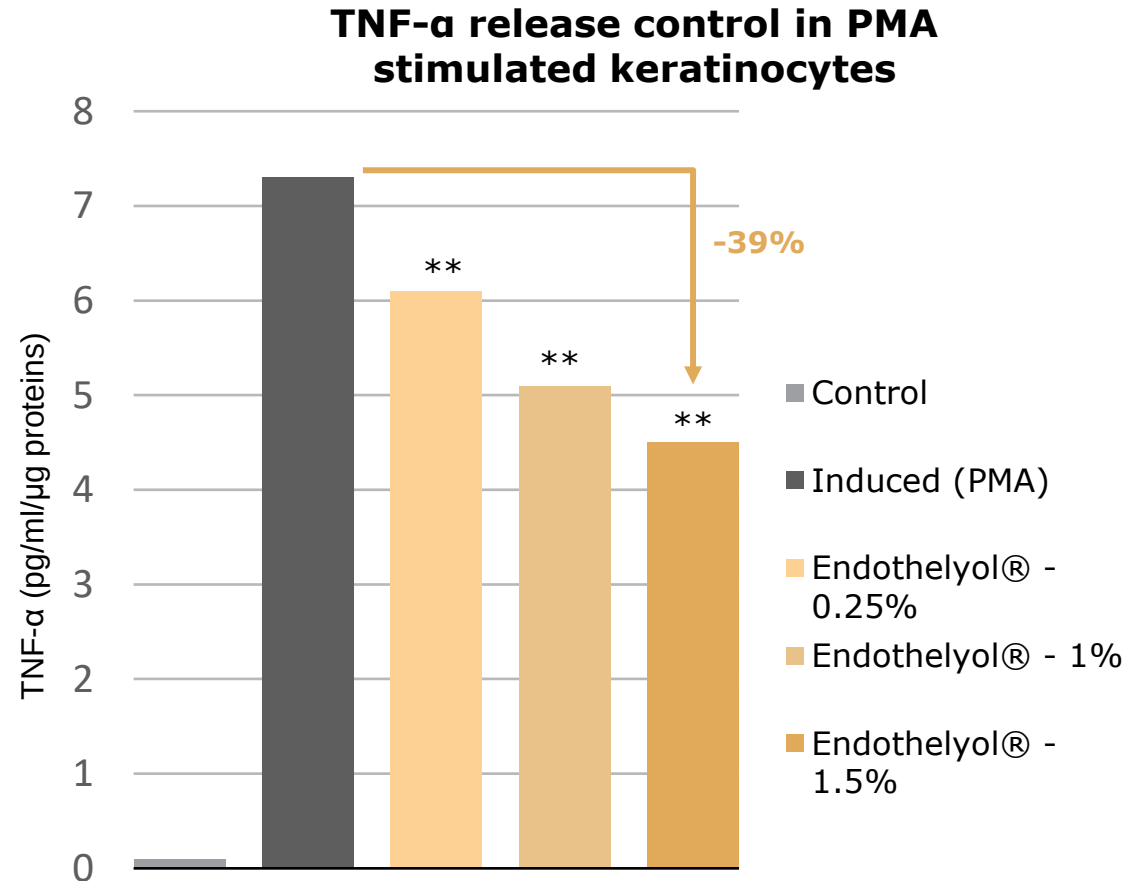
Tight control of redness factors

TNF- α release control in PMA stimulated keratinocytes

PROTOCOL

Endothelyol® was tested for its ability to control redness factors. ELISA quantification of TNF- α released was done on keratinocytes extracted from clinical human skin sample incubated with the product for 48h.

→ Endothelyol®
downregulates TNF- α
release by -39%.



**p<0.01 compared to stimulated control, Student's t Test

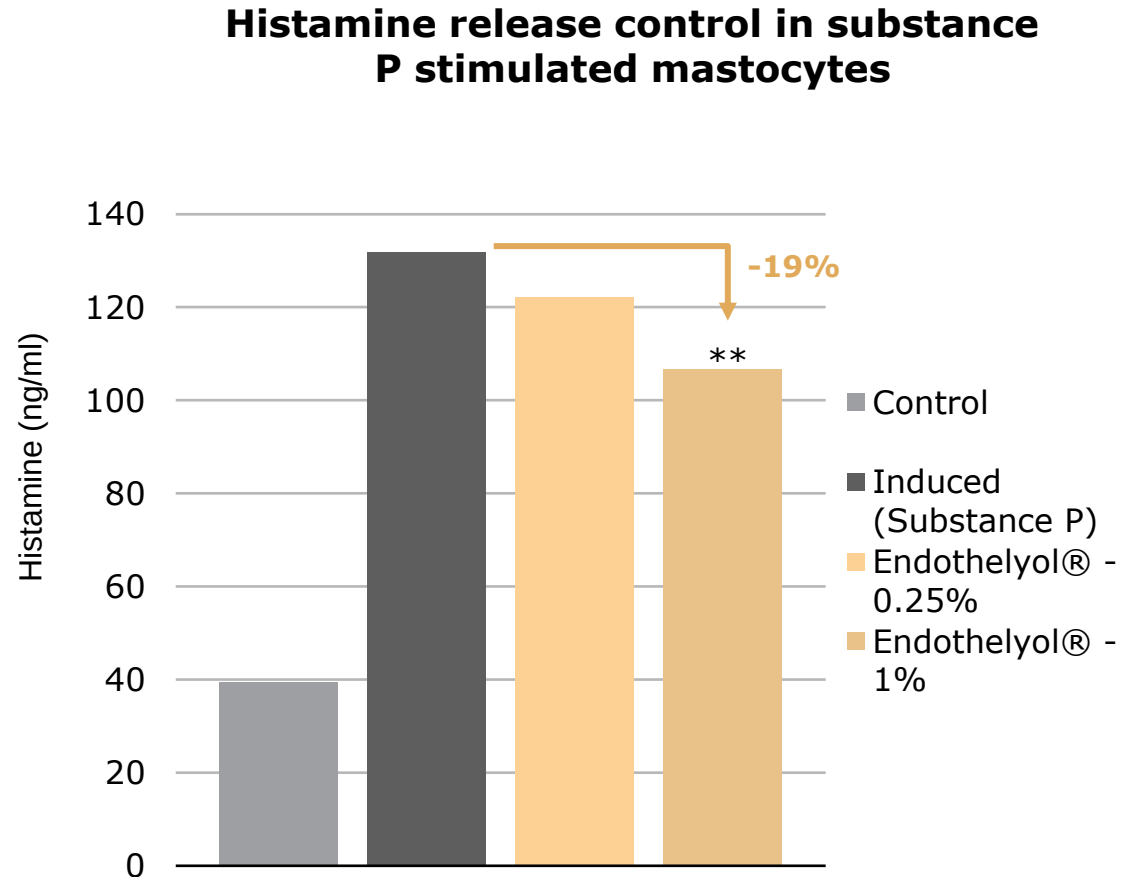
Tight control of redness factors

Histamine release control in substance P stimulated mastocytes

PROTOCOL

Endothelyol® was tested for its ability to control itching. ELISA quantification of histamine released was done on rodent mastocytes incubated with the product for 30 min.

→ Endothelyol®
downregulates histamine
release by -19%.



**p<0.01 compared to stimulated control, Student's t Test

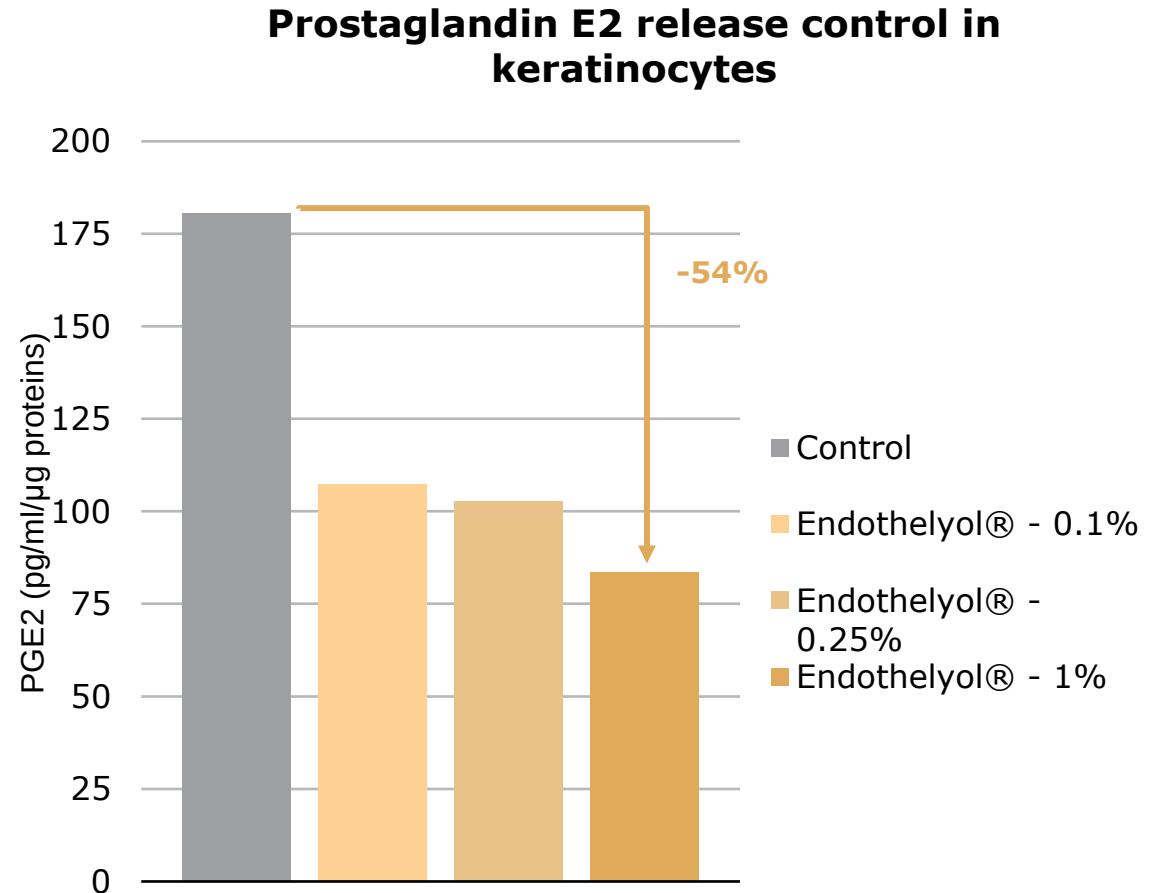
Tight control of redness factors

Prostaglandin E2 release control in keratinocytes

PROTOCOL

Endothelyol® was tested for its ability to control vasodilation. ELISA quantification of PGE2 released was done on keratinocytes extracted from clinical human skin sample incubated with the product for 48h.

→ Endothelyol®
downregulates PGE2
release by -54%.



Tight control of redness factors

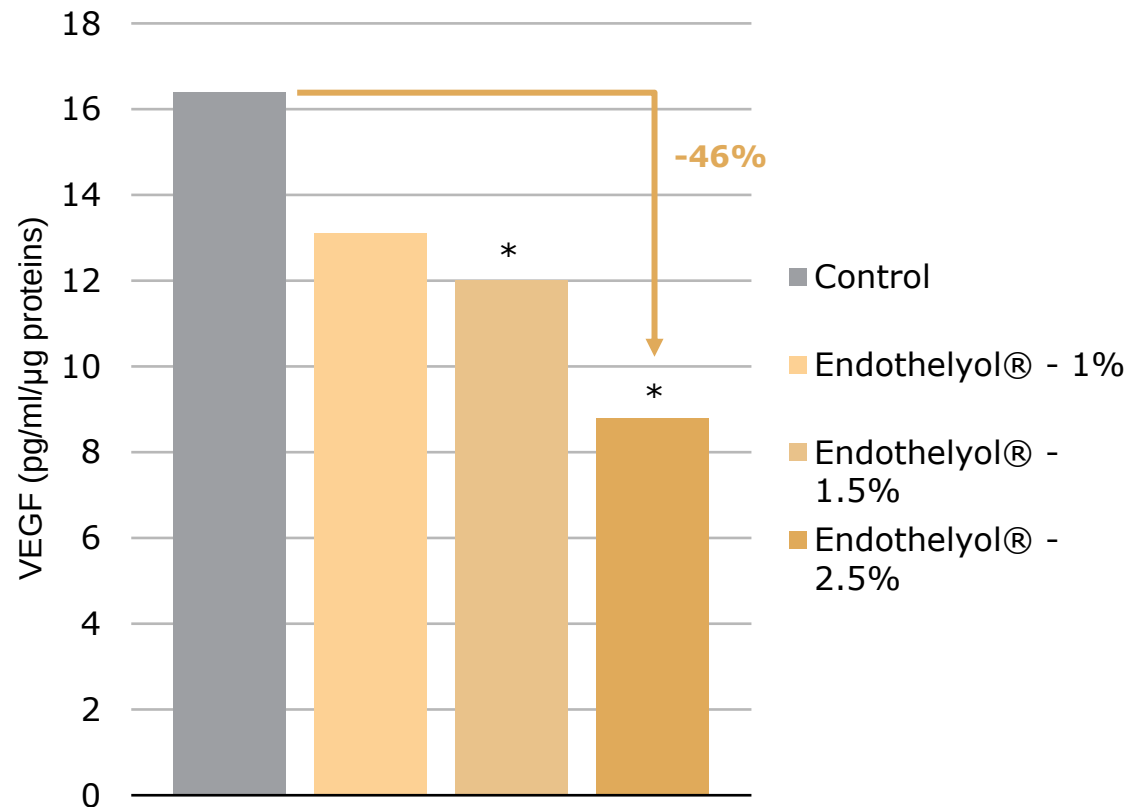
Vascular Endothelyol Growth Factor release control in keratinocytes

PROTOCOL

Endothelyol® was tested for its ability to control vasodilation. ELISA quantification of PGE2 released was done on keratinocytes extracted from clinical human skin sample incubated with the product for 48h.

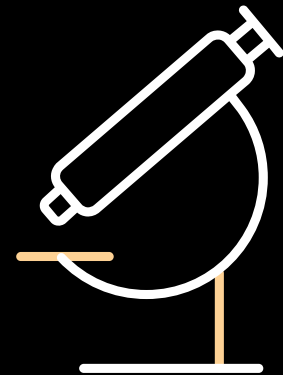
→ Endothelyol® reduces formation of new blood vessels by 46%.

Vascular Endothelyol Growth Factor release control in keratinocytes



*p<0.05 compared to stimulated control, Student's t Test

ex vivo evaluations



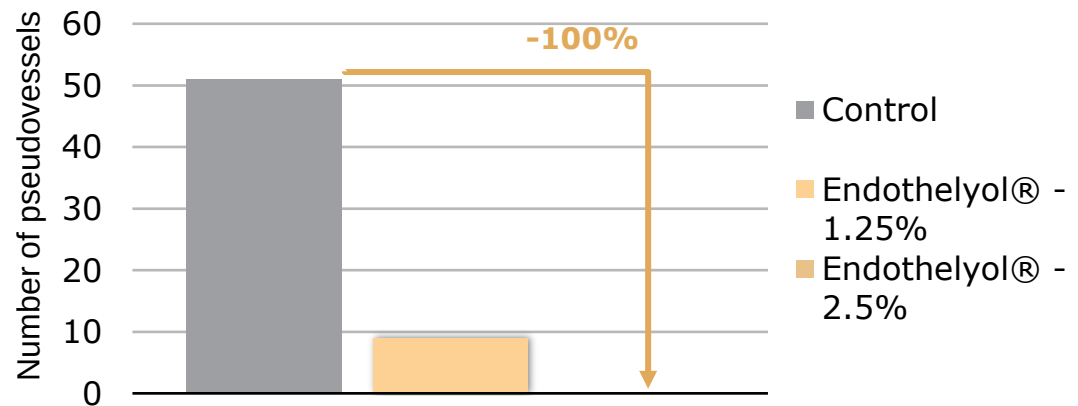
Inhibition of neo-vessels growth

Quantification of neo-vessels

PROTOCOL

Endothelyol® was tested for its anti-angiogenesis properties. A quantification of the number of neo-vessels created upon time was done on HskMEC (Microvascular endothelial human skin cells) cultured on MATRIGEL™ with or without the product for 17h.

Quantification of neo-vessels



Dashed line : contour of a neo-vessel

→ **Endothelyol® blocks neo-vessels growth.**

* $p < 0.05$ compared to stimulated control, Student's t Test

Clinical assessment



Reduction of rosacea

Clinical study #1

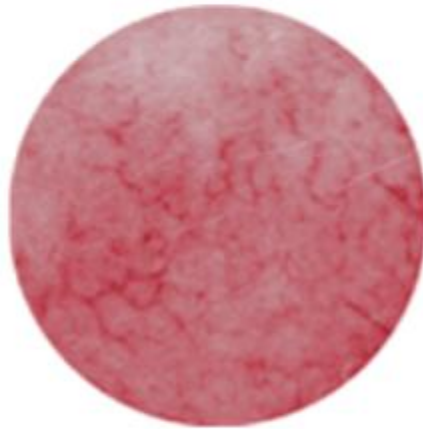
Protocol:

- ❖ Double blind *versus* placebo.
- ❖ **20 volunteers**
- ❖ Usage: A placebo and a cream containing 2% Endothelyol® were applied for 28 days, twice a day, on either half of the face.
- ❖ Evaluation: The rosacea on the face was evaluated under dermatological control using heamoglobin measurement (SiaScope).



Reduction of rosacea

Day 0



Day 28



Endothelyol[®] 2%

❖ **After 28 days, Endothelyol[®] reduces rosacea by 7% to 12%.**

Improvement of skin brightness

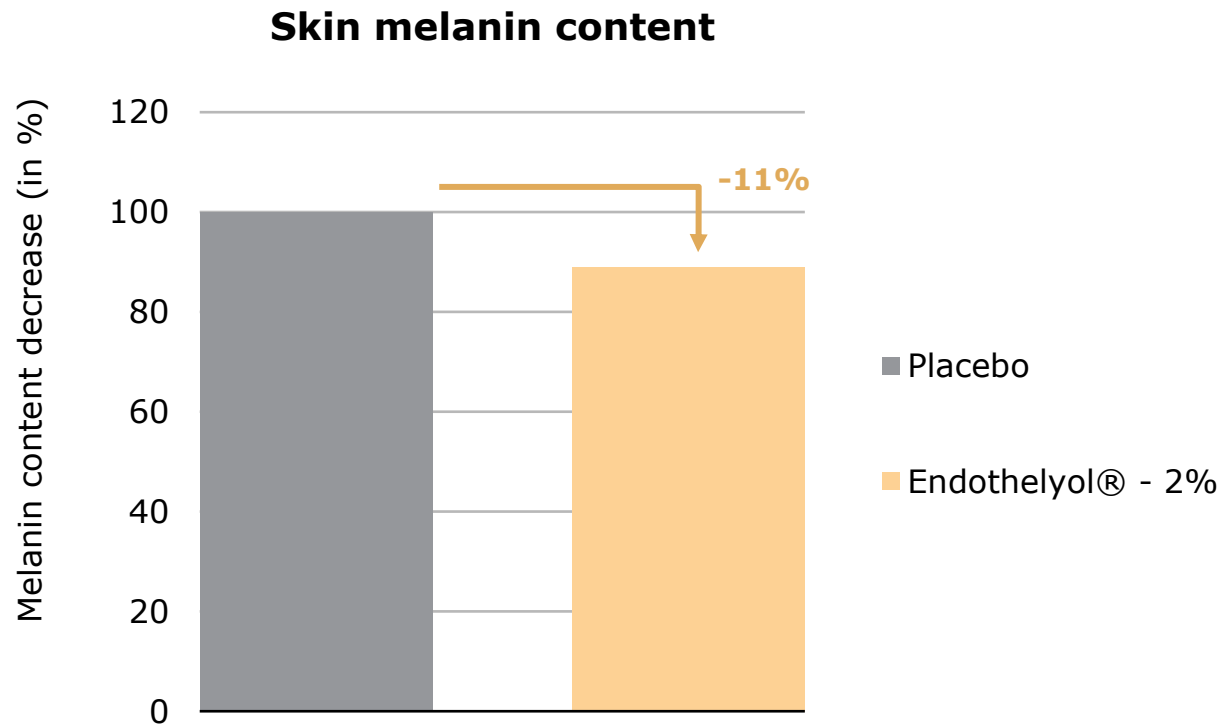
Clinical study #2

Protocol:

- ❖ Double blind *versus* placebo.
- ❖ **20 volunteers**
- ❖ Usage: A placebo and a cream containing 2% Endothelyol® were applied for 28 days, twice a day, on either half of the face.
- ❖ Evaluation: The quantification of skin melanin content was done by Siascopy™

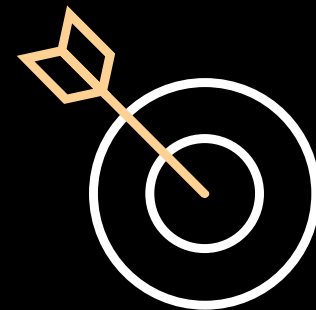


Improvement of skin brightness



After 28 days, Endothelyol® improves skin color: 11% of brightening.

Summary



Technical information

Endothelyol™

INCI

Water (and) Glycerin (and) Rosmarinyl Glucoside (and) Caffeyl Glucoside (and) Gallyl Glucoside

Origin

Biotechnology

Preservation

Preservative Free

Appearance

Clear amber solution

Solubility

Water soluble

Dosage

0.25% to 2.5%

Processing

Can be incorporated in any formula in liquid form at pH below 6.5 and temperature below 45°C

Compliance



Thank you

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